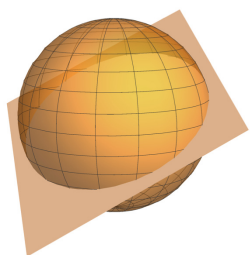


Lecture 23 Stokes' Theorem Cont'd

Recap: $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds.$
 $\uparrow$
 Orientable, simply-connected

Line integral $\rightarrow$ surface integral.

Example. Let C be the curve of intersection of the plane $Ax + By + Cz = 0$ and the sphere $x^2 + y^2 + z^2 = a^2$. Show that $\oint_C y \, dx + z \, dy + x \, dz = \pm \frac{2a^2(A+B+C)}{\sqrt{A^2+B^2+C^2}}.$



Sol. $\vec{F} = \langle y, z, x \rangle$ $\oint_C \vec{F} \cdot d\vec{r}.$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = -\vec{i} - \vec{j} - \vec{k}.$$

S : the disk region enclosed by C on the plane.

The unit normal vector of S is

$$\vec{n} = \pm \frac{\langle A, B, C \rangle}{\sqrt{A^2 + B^2 + C^2}}$$

$$(\nabla \times \vec{F}) \cdot \vec{n} = \pm (-\vec{i} - \vec{j} - \vec{k}) \cdot \frac{\langle A, B, C \rangle}{\sqrt{A^2 + B^2 + C^2}} = \pm \frac{A+B+C}{\sqrt{A^2 + B^2 + C^2}}$$

By Stokes' Theorem,

$$\oint \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$= \pm \frac{A+B+C}{\sqrt{A^2+B^2+C^2}} \underbrace{\iint_S ds}_{\text{area of } S}$$

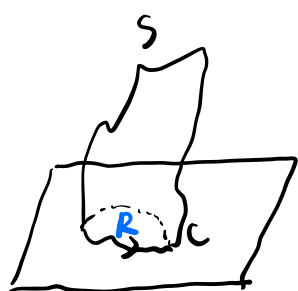
$$= \pm \frac{2a^2(A+B+C)}{\sqrt{A^2+B^2+C^2}}. \quad \square$$

Surface integral $\rightarrow$ line integral.

$$\vec{F} = \nabla \times \vec{A}.$$

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds = \oint_C \vec{A} \cdot d\vec{r}.$$

Example. Let C be an arbitrary simple closed curve on the xy -plane in the xyz -space, and S be an arbitrary surface above the xy -plane with boundary C . 1. Verify that $\vec{i} = \nabla \times (-\frac{z}{2}\vec{j} + \frac{y}{2}\vec{k})$



2. Show that $\iint_S \vec{i} \cdot \vec{n} \, ds = 0$.

$$\begin{aligned} \text{Sol. 1. } \nabla \times (-\frac{z}{2}\vec{j} + \frac{y}{2}\vec{k}) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & -\frac{z}{2} & \frac{y}{2} \end{vmatrix} \\ &= \left(\frac{\partial}{\partial y}(\frac{y}{2}) - \frac{\partial}{\partial z}(-\frac{z}{2}) \right) \vec{i} \\ &= \vec{i}. \end{aligned}$$

$$2. \vec{A} = -\frac{z}{2}\vec{j} + \frac{y}{2}\vec{k}$$

$$\vec{i} = \nabla \times \vec{A}.$$

$$\iint_S \vec{i} \cdot \vec{n} \, ds = \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds$$

$$= \oint_C \vec{A} \cdot d\vec{r}$$

By Stokes' Theorem.

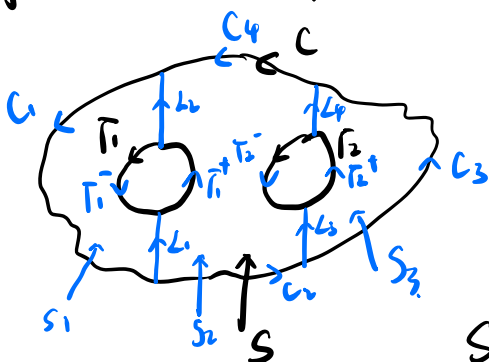
$$= \iint_R (\nabla \times \vec{A}) \cdot \vec{k} \, dA.$$

By Green's / Stokes' Theorem.

$$= \iint_R 0 \, dA$$

$$= 0. \quad \square.$$

Surfaces with multiple boundaries.



$$C = C_1 + C_2 + C_3 + C_4$$

$$\Gamma_1 = \Gamma_1^- + \Gamma_1^+$$

$$\Gamma_2 = \Gamma_2^- + \Gamma_2^+$$

$$S_1: C_1 + L_1 - \Gamma_1^- + L_2$$

$$S_2: C_4 - L_2 - \Gamma_1^+ - L_1 + C_2 + L_3 - \Gamma_2^- + L_4$$

$$S_3: C_3 - L_4 - \Gamma_2^+ - L_3.$$

By Stokes' Theorem for simply-connected surfaces:

$$\left(\int_{C_1} + \int_{L_1} - \int_{\Gamma_1^-} + \int_{L_2} \right) \vec{F} \cdot d\vec{r} = \iint_{S_1} (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$\left(\int_{C_4} - \int_{L_2} - \int_{\Gamma_1^+} - \int_{L_1} + \int_{C_2} + \int_{L_3} - \int_{\Gamma_2^-} + \int_{L_4} \right) \vec{F} \cdot d\vec{r} = \iint_{S_2} (\nabla \times \vec{F}) \cdot \vec{n} \, ds$$

$$\left(\int_{C_3} - \int_{L_4} - \int_{\Gamma_2^+} - \int_{L_3} \right) \vec{F} \cdot d\vec{r} = \iint_{S_3} (\nabla \times \vec{F}) \cdot \vec{n} \, ds.$$

$$\int_C \vec{F} \cdot d\vec{r} - \int_{\Gamma_1} \vec{F} \cdot d\vec{r} - \int_{\Gamma_2} \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds.$$

Theorem (Stokes' Theorem for General Surfaces) Let S be an orientable surface in $\mathbb{R}^3$ with n holes. Denote C to be its outer boundary and $\Gamma_1, \Gamma_2, \dots, \Gamma_n$ to be its inner boundaries. Suppose $\vec{F}$ is defined and C' on S , then $\oint_C \vec{F} \cdot d\vec{r} - \sum_{i=1}^n \oint_{\Gamma_i} \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds$. Here $\vec{n}$ is the unit normal vector to S with orientation determined by the right-hand grab rule applied to the outer boundary C .

Divergence.

$$\text{Operator } \nabla = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}.$$

$$\text{Gradient of } f : \nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}.$$

$$\text{Curl of } \vec{F} : \vec{F} = \langle F_x, F_y, F_z \rangle.$$

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \\ &= \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{i} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \vec{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \vec{k}. \end{aligned}$$

$$\begin{aligned} \text{Divergence of } \vec{F} : \nabla \cdot \vec{F} &= \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle \cdot \langle F_x, F_y, F_z \rangle \\ &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \end{aligned}$$

Example. Compute the divergence of the vector field

$$\vec{F} = \frac{\vec{r}}{|\vec{r}|^p} = \frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{p/2}}, \text{ where } p > 0 \text{ is a real number.}$$

$$\text{Sol. } \frac{\partial F_x}{\partial x} = \frac{\partial}{\partial x} \left(\frac{x}{(x^2 + y^2 + z^2)^{p/2}} \right)$$

$$= \frac{(x^2 + y^2 + z^2)^{p/2} - x \cdot 2x \cdot \frac{p}{2} (x^2 + y^2 + z^2)^{p/2 - 1}}{(x^2 + y^2 + z^2)^p}$$

$$= \frac{|\vec{r}|^p - p x^2 |\vec{r}|^{p-2}}{|\vec{r}|^{2p}}$$

$$\frac{\partial F_y}{\partial y} = \frac{|\vec{r}|^p - p y^2 |\vec{r}|^{p-2}}{|\vec{r}|^{2p}}$$

$$\frac{\partial F_z}{\partial z} = \frac{|\vec{r}|^p - p z^2 |\vec{r}|^{p-2}}{|\vec{r}|^{2p}}$$

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = \frac{3|\vec{r}|^p - p(x^2 + y^2 + z^2)|\vec{r}|^{p-2}}{|\vec{r}|^{2p}} = \frac{3-p}{|\vec{r}|^p}$$

Remark: When $p=3$, $\vec{F}$ is the gravitational/electric vector field.

$$\nabla \cdot \vec{F} = 0$$

$$\nabla \times \vec{F} = \vec{0}$$

Properties.

$$1. \nabla \cdot (\vec{F} + \vec{G}) = \nabla \cdot \vec{F} + \nabla \cdot \vec{G}$$

$$2. \nabla \cdot (c\vec{F}) = c \nabla \cdot \vec{F}$$

$$3. \nabla \times (\vec{F} + \vec{G}) = \nabla \times \vec{F} + \nabla \times \vec{G}$$

$$4. \nabla \times (c\vec{F}) = c \nabla \times \vec{F}.$$

$$\text{Curl of the gradient} \quad \nabla \times \nabla f = \vec{0}$$

$$\text{Divergence of the curl} \quad \nabla \cdot (\nabla \times \vec{F}) = 0$$

$$\text{Divergence of the gradient} \quad \nabla \cdot \nabla f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Laplacian operator